

Autosomal dominant disorders to illustrate genetic principles

GNET DLA_Before Mendelian Inheritance: AD and AR

Three parts:

- A. Reading assignment
- B. Autosomal dominant**
- C. Autosomal recessive

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Inheritance: AD and AR – *DLAs*

SOM.MK.III.BPM1.1.FTM.3.GNET.0020	Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity
SOM.MK.III.BPM1.1.FTM.3.GNET.0021	From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
SOM.MK.III.BPM1.1.FTM.3.GNET.0022	Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families
SOM.MK.III.BPM1.1.FTM.3.GNET.0023	Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule
SOM.MK.III.BPM1.1.FTM.3.GNET.0024	Describe and differentiate between pseudoautosomal dominance, co-dominance, and incomplete dominance; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0025	Describe co-dominance, giving examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0026	Describe incomplete dominance, giving examples

Inheritance: AD and AR - *Flipped classroom lecture*

SOM.MK.III.BPM1.1.FTM.3.GNET.0027	Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles
SOM.MK.III.BPM1.1.FTM.3.GNET.0028	Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence
SOM.MK.III.BPM1.1.FTM.3.GNET.0029	Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD
SOM.MK.III.BPM1.1.FTM.3.GNET.0030	Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance
SOM.MK.III.BPM1.1.FTM.3.GNET.0031	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria)
SOM.MK.III.BPM1.1.FTM.3.GNET.0032	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria
SOM.MK.III.BPM1.1.FTM.3.GNET.0033	Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0034	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0035	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0036	Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations

Autosomal dominant disorders

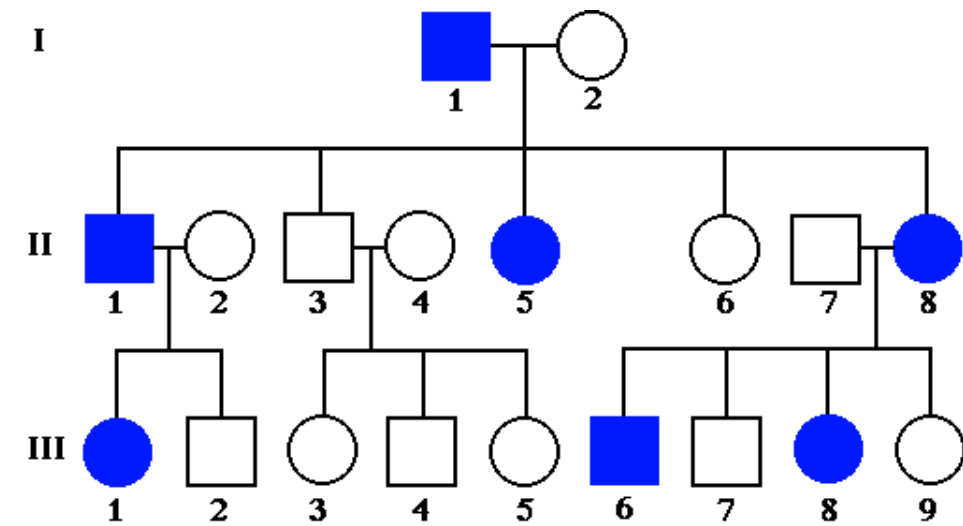
- In this brief video (and throughout the course), examples of human genetic disorders will be presented to illustrate **genetic principles**
- As we learn more about the disorders, and human genetics, the principles that they may illustrate will grow
- DLA 3 Parts:
 - A. Reading assignment
 - B. Autosomal dominant**
 - C. Autosomal recessive

AD disorders: we expect you to remember these examples, their names, their mode of transmission, their main clinical features, and when listed here, the name of the gene (gene name abbreviations in *italics*)

- Familial hypercholesterolemia (due to deficiency of *LDLR*)
 - Example of haploinsufficiency; Allelic heterogeneity and locus heterogeneity
 - May be strongly influenced by lifestyle factors
- Huntington disease (gene name is huntingtin; *HTT*)
 - Example of gain of function of the protein
 - Triplet repeat expansion disorder, displays anticipation
 - High penetrance, 90% of cases are adult onset (so age dependent penetrance)
- Myotonic dystrophy
 - Also gain of function, but at the mRNA level
 - Triplet repeat expansion disorder, anticipation, pleiotropy
- Marfan syndrome (*FBN1*)
 - Has haploinsufficiency forms, and dominant negative forms (see page 45 & 46 of Korf)
- Osteogenesis imperfecta (*COL1A1*, and *COL1A2*)
 - Brittle bone disease
 - Has haploinsufficiency forms, and dominant negative forms (see page 45 & 46 & 47 of Korf)
- Achondroplasia (*FGFR3*)
 - Example of Gain of function, unregulated tyrosine kinase signaling
 - Almost 100% penetrant; Mutation hot spot (new mutation)
- Neurofibromatosis type I (*NF1*)
 - Example of AD because of loss of heterozygosity, *NF1* as a tumor suppressor gene
 - Example of disorder which is highly penetrant, but has variable expressivity; mutation hot spot, high allelic heterogeneity
- Acute intermittent porphyria (abbreviated as AIP, no need to memorize a gene name for this one)
 - Heme synthesis deficiency; AIP is a rare example of an enzyme deficiency that is autosomal dominant; most enzyme deficiencies are recessive.
 - AIP is dominant because of high cellular demand for heme

As we progress through BPM1 and 2, and then into year 2 we will return to many of these disorders as we illustrate other genetic principles

Autosomal dominant



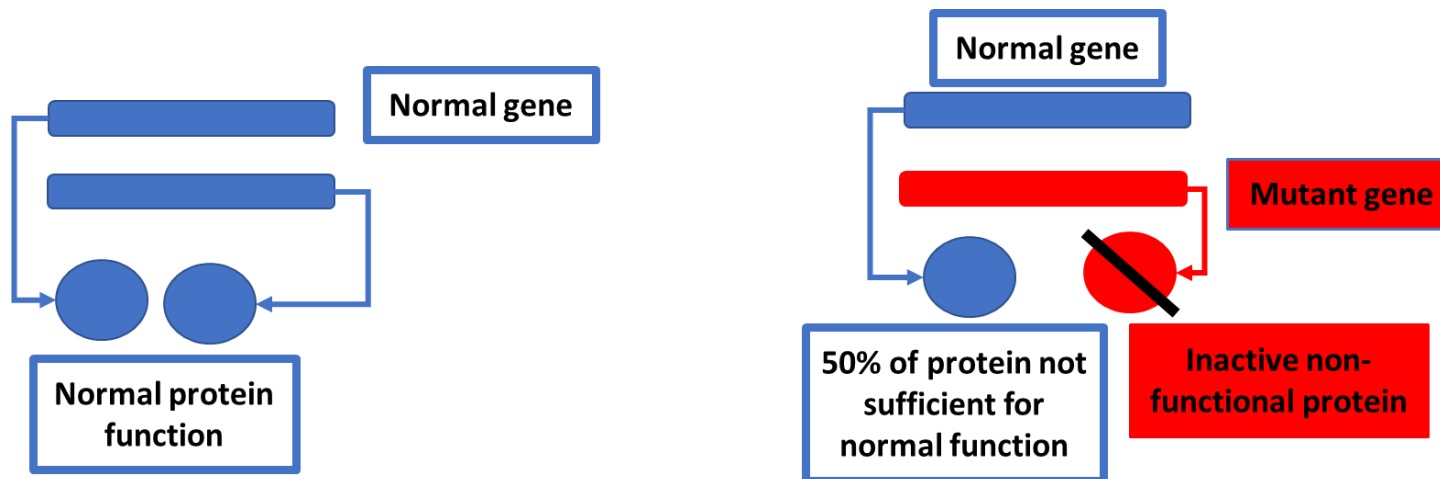
- Autosomal dominant disorders are generally **heterozygous**
- Affected individuals generally have one copy of the pathogenic allele and one normal copy of the gene (AA*)
 - **Assumption:** Heterozygosity for autosomal dominant disorders, unless otherwise specified
- Homozygosity for the pathogenic allele is rare; and generally, more severe phenotype

Genetic (molecular) mechanisms which explain autosomal dominant inheritance

1. Haploinsufficiency
2. Gain of function
3. Dominant negative
4. Loss of heterozygosity
 - Inheritance of one disease allele in the germline occurs (from the gamete), and then loss of the second, functional copy occurs, either during fetal development, or following birth sometime during the lifetime of the individual. We give only one example in this section (*NF1*), but will develop it in much more detail during the Cancer Genetics section of the course

Examples of autosomal dominant: **Haploinsufficiency**

- Heterozygosity for the mutant allele of gene
- The normal allele produces the functional protein
- No protein production from the mutant allele (No functional protein from mutant allele)
- 50% of the functional protein produced → NOT sufficient for the physiological function of the protein → Disease phenotype



Familial hypercholesterolemia: Haploinsufficiency

- LDL receptor (*LDLR*) is found on the hepatocyte (liver cell) and all cells
- LDL receptor binds to LDL particles to clear them from the blood
- Heterozygosity for loss of function mutation in one *LDLR* allele → **fewer LDL receptors (50%)** → Higher LDL (High serum cholesterol)
- Xanthomas: Painless, yellowish deposits of cholesterol on tendons, elbows, ankles, wrists, palpebral areas (xanthelasma)
- High serum cholesterol → Atherosclerosis and cardiovascular disease

Xanthoma: Cholesterol depositions on tendons and knee



Xanthelasma: Cholesterol depositions on the eyelids



Familial hypercholesterolemia (*LDLR* deficiency): **Allelic heterogeneity**

- Numerous different mutation alleles in *LDLR* may destroy the function of a gene
 - **This is the definition of allelic heterogeneity**
- The *LDLR* pathogenic variants exhibit a high degree of allelic heterogeneity
 - Many different pathogenic alleles are known
 - In fact, thousands of different *LDLR* pathogenic variants are known, each one is a different allele
 - Same gene (*LDLR* gene) but different types of mutations

Familial hypercholesterolemia: **Locus heterogeneity**

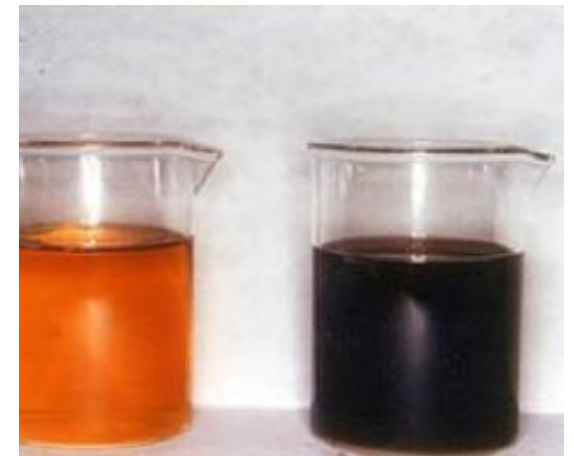
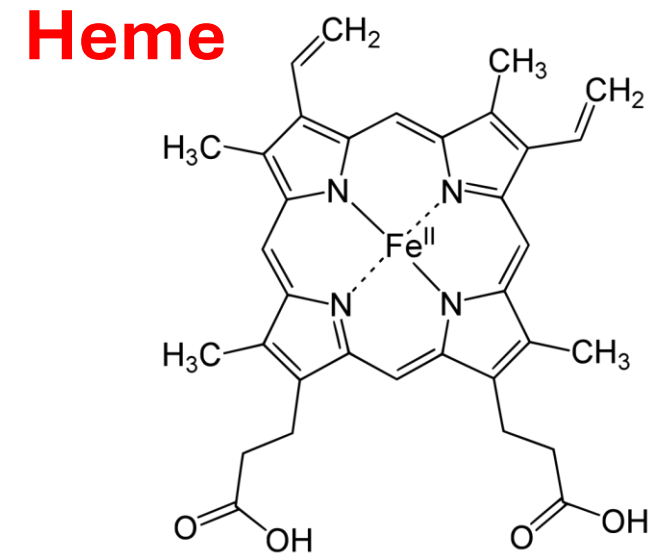
- Hypercholesterolemia can be caused by mutations in other genes
- So, a person could have FHC **and** have two functional copies of the *LDLR* gene.
 - Meaning the FHC in this family ***is not*** caused by *LDLR* mutation
 - So, the FHC in this person is caused by mutation somewhere else (*APOB* gene)
- This is an example of **locus heterogeneity**
 - Single gene disorder caused by mutations in different genes
 - FH caused by *LDLR* (most common) or *APOB* (less common) gene

Acute intermittent porphyria: **Haploinsufficiency of an enzyme**

- Enzymes are catalysts, hence low levels of an enzyme satisfies the needs of the cell and for the person to remain disease free.
- In most cases, heterozygosity for a loss of function mutation in just about ANY enzyme would not have a phenotype (no disease)
 - Only one functional copy of that gene sufficient for physiological function
- A notable **exception**: **Acute intermittent porphyria** and some other porphyrias

Acute intermittent porphyria

- A lot of heme is needed by the body
- Heme is needed in:
 1. Red blood cells (hemoglobin)
 2. Any cell with a mitochondria
 3. Liver cells for cytochromes (detoxification enzymes)
- Since there is **so much demand for heme**, both alleles must be functional
- Intermittent symptoms of anxiety, nausea, severe abdominal pain, constipation, diarrhea; Purple red urine
 - Due to accumulation of precursors of heme synthesis
- Patient history: <https://www.youtube.com/watch?v=SM6BFzum8yl>



Dark urine on exposure to air and light

Examples of autosomal dominant: **Gain of function**

- Two types of gain of function:
- Mutation causes the gene to do more of what it normally does
 - Can be caused by a variety of reasons:
 1. Maybe **more protein** is synthesized (generally means more activity) or
 2. Protein acts in an **unregulated manner**
- Mutation causes the gene to do a “new” function
 - Same as saying “attainment of a novel function”

Achondroplasia

- A specific DNA base pair change causes the encoded protein to function in an **unregulated** manner.
- *FGFR3* gene encodes a receptor protein **tyrosine kinase**
- Normally FGFR3 protein signals to bone growth plates during development to begin ossification
- If FGFR3 signals in an unregulated manner, **premature ossification** of bone growth plates occurs, and person develops disproportionate short stature
 - Shortening of long bones
 - Facial dysmorphology

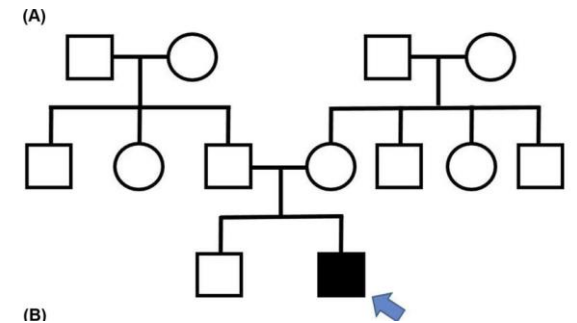


Source: <https://showhope.org/2011/11/16/overcome-with-emotion/>



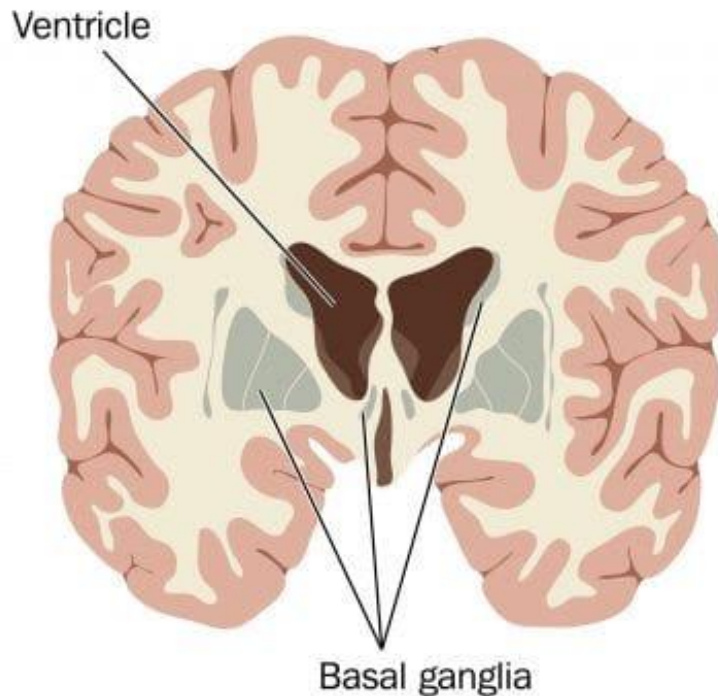
A very specific mutation causes FGFR3 protein to become unregulated

- Point mutation in gene → Amino acid substitution mutation in protein
- Individuals with achondroplasia have the **same pathogenic variant**
 - Glycine 380 → Arginine amino acid substitution (Gly380Arg)
 - Gly at position 380 of the protein replaced by Arg (*no need to memorize*)
 - There is **very little** allelic heterogeneity
- Homozygosity for FGFR3 (not compatible with life)
- “Mutation hotspot” for ***denovo* (new) mutation**
 - A child with achondroplasia with no family history

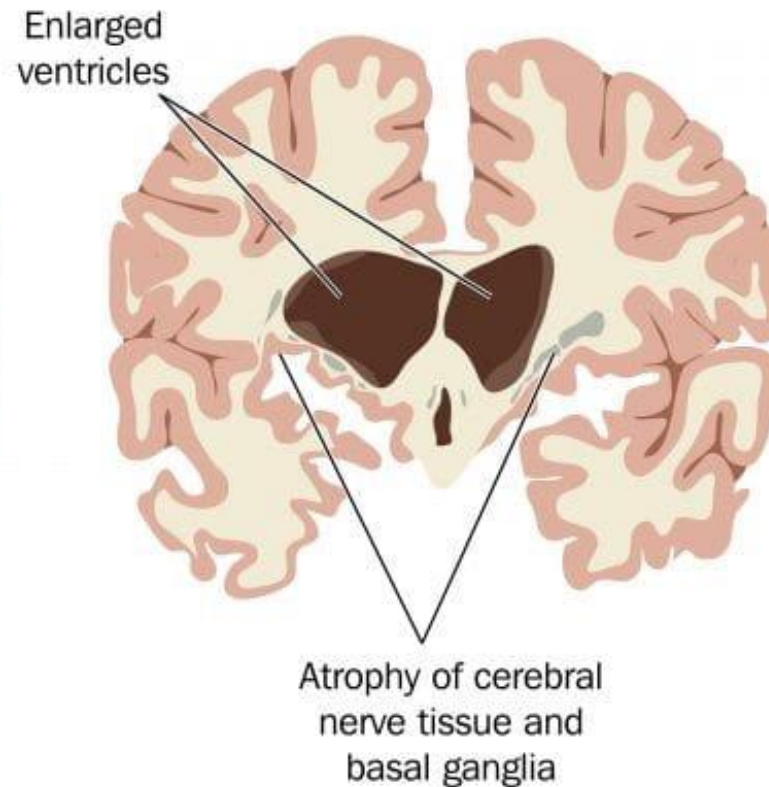


Huntington disease: Gain of function; “Attainment of novel function”

Normal brain



Huntington's disease



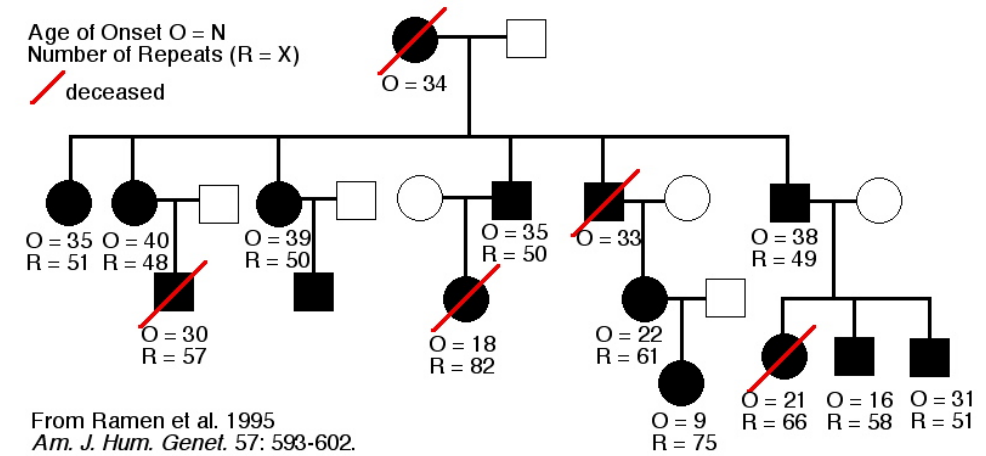
- Neurological disorder
- Basal ganglia affected
- Involuntary movements, chorea, rigidity, cognitive decline
- Age of onset: 30-50 years

Huntington disease: Gain of function

- Caused by an **expansion of an unstable triplet** of bases in an **exon** of the **huntingtin gene** (*HTT* gene)
- Since expansion is an **exon**, the triplets are **codons** (CAG repeat), so they will be translated into the encoded protein
- Leads to an expanded tract of **glutamine** amino acids in the encoded protein (polyglutamine tract expansion)
- Causes new function(s) of the protein (unknown) → Progressive death of neurons
- Huntingtin protein accumulates in neurons → Neuronal death

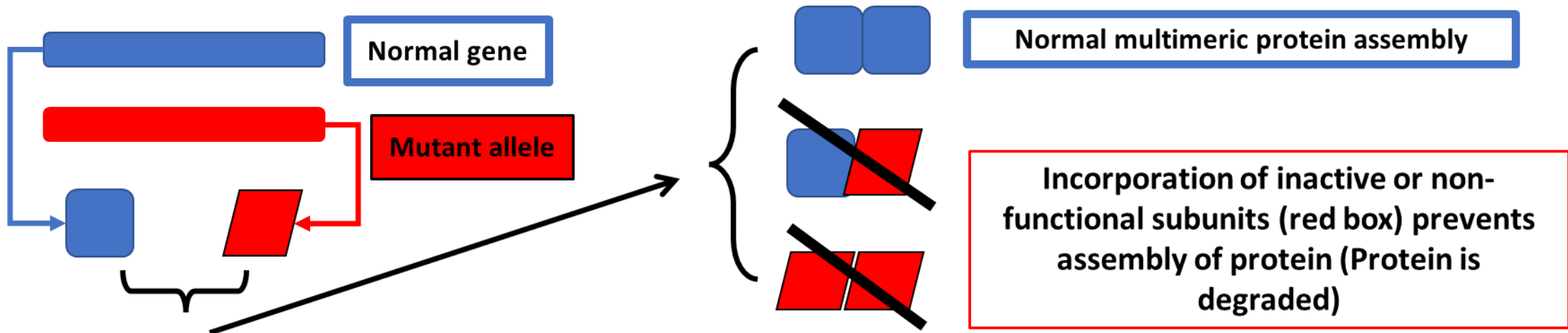
Huntington disease: Anticipation

- As the repeat expands, it tends to (but not always) become more unstable.
- So as the gene is passed down generations, the repeat length might increase (longer polyglutamine tracts)
- Number of repeats: Correlate with severe and earlier onset of the disorder
- Called “**Anticipation**”

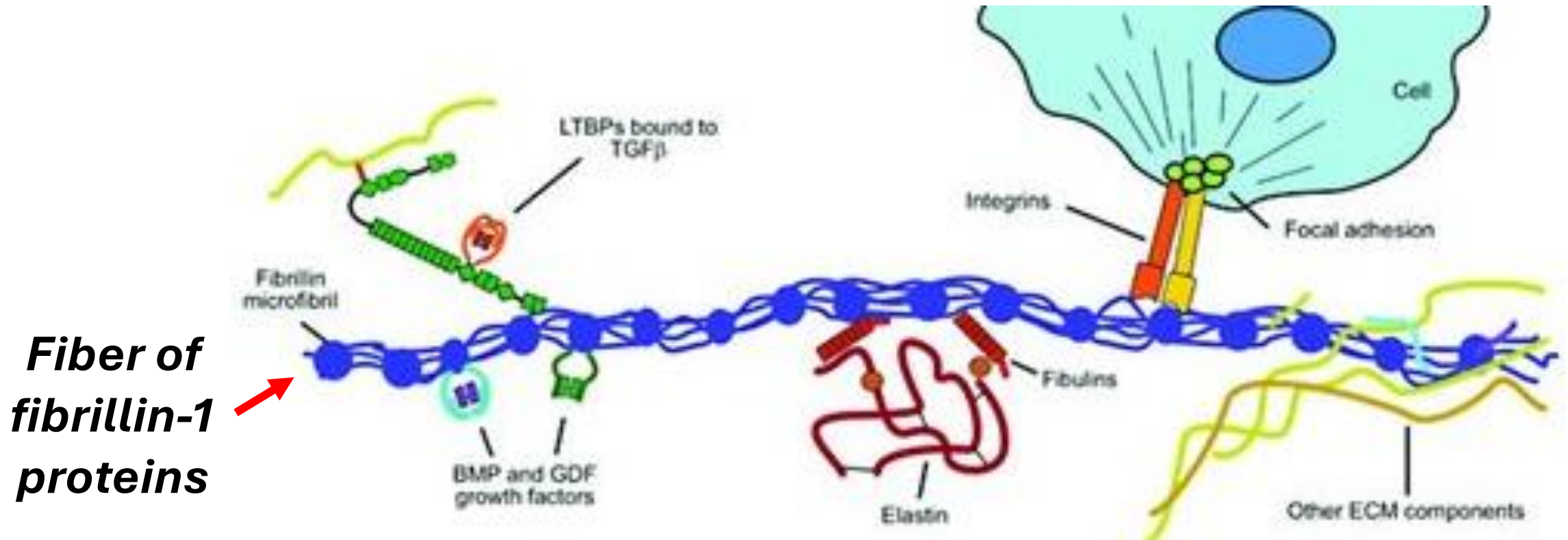


Example of autosomal dominant: “**Dominant negative**” effect

- Usually observed in **structural protein disorders**
 - Marfan syndrome and osteogenesis imperfecta
- Multiple subunit assembly is required for the protein function
- Presence of a mutant protein prevents multimer formation (Poisons the system)



Marfan syndrome: mutation of the *FBN1* gene which encodes a connective tissue protein called fibrillin-1



Many other connective tissue proteins and proteins of the extracellular matrix interact with fibrillin

Marfan syndrome: Genotype-phenotype correlation

- 1. Haploinsufficiency:** Mutations that lead to less fibrillin-1 protein. *FBN1* loss of function mutation, and the less severe phenotype (50% functional fibrillin-1)
- 2. Dominant negative:** A mutation that allows the fibrillin protein to be made in normal amounts, but what is made will interact inappropriately with the normally formed fibrillin-1 (from the other allele) and also other connective tissue proteins (**dominant negative**). More severe phenotype (less than 50% functional fibrillin-1)

Osteogenesis imperfecta (brittle bone disease)

- Caused by mutation in **collagen gene**
- Caused by mutation of *COL1A1* or *COL1A2* (Locus heterogeneity)
- Autosomal dominant
- Frequent fractures, blue sclera
- **Haploinsufficiency** mutations (less severe)
 - Less collagen formed
- **Dominant negative mutations** (more severe)
 - Presence of abnormal/distorted collagen chain (mutant allele) prevents assembly of collagen



Mikael Häggström, M.D. - Author info - Reusing imagesWritten informed consent was obtained from a parent of the individual, including for online publication. / CC0

Dominant negative vs haploinsufficiency

- When considering Marfan syndrome, and OI, the disorder is likely to be more severe when caused by the **dominant negative mutation**

Haploinsufficiency = Less severe = half the Normal amount



Dominant negative variant = More severe =
Very little functional protein



Neurofibromatosis

- Caused by inheritance of one pathogenic *NF1* allele
 - About ½ of cases are inherited from an affected parent
 - Approximately ½ of cases due to **de-novo mutation** during gametogenesis (Parent not affected; no family history; Mutation hotspots in NF1)
 - Heterozygous and cells have one functional NF1 allele
- During the lifetime, a second mutation may cause loss of the remaining functional *NF1* allele in a somatic cell (Loss of heterozygosity) → The cell now has no functional NF1 protein
 - Almost guaranteed that this will happen (usually many times)
 - Cells with no functional NF1; these grow into the **tumors**

Function of NF1

- The NF1 protein suppresses cellular proliferation
 - *We will discuss NF1 in much more detail later after the first exam*
- Loss of NF1 leads to formation of benign tumors (neurofibromas), and also increased risk for other cancer types
- NF1 has very **high penetrance**
 - Individuals with the mutation have the phenotype/ clinical features
- NF1 shows **variable expression**
 - Signs and symptoms may be highly variable among different individuals, even within the same family
 - Some are severely affected, and others less severely affected

Neurofibromas, are usually noncancerous (benign) tumors that are often located on or just under the skin

Credit: [GeneReviews, University of Washington](#)
[Use of illustrations and other content](#)



Café au lait spots

Credit: [GeneReviews, University of Washington](#) Use of illustrations and other content



Lisch nodules of the eye

<https://emedicine.medscape.com/article/1219222-overview>

Summary:

- Calculation of recurrence risks and recognition of pedigrees
- Molecular mechanisms in autosomal dominant disorders
 - Haploinsufficiency
 - Gain of function
 - Dominant negative effect
 - New mutation
- Clinical features and genetic mechanisms of the disorders listed

Please email
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you have any questions

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- Phylloerythrin (PHE)
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- Marfan syndrome (FBN1)
 - Mainly haploinsufficiency for FBN1, and dominant negative forms (see page 25 & 26 of 2017)
- Osteogenesis imperfecta (COL1A1, and COL1A2)
 - Brittle bone disease
 - Mainly haploinsufficiency for FBN1, and dominant negative forms (see page 25 & 26 of 2017)
- Achromatopsia (CERN)
 - Example of loss of function, unregulated by white noise signaling
 - Almost 100% penetrant; Mutations hot spot (see mutation)
- Neurofibromatosis type 1 (NF1)
 - Example of loss of function of loss of heterozygosity; NF1 as a tumor suppressor gene
 - Example of disorder which is highly penetrant, but has variable expressivity; mutation hot spot; high allele heterogeneity
- Acute intermittent porphyria (abbreviated as AIP, no need to memorize a gene name for this one)
 - Heme synthesis deficiency; AIP is a rare example of an enzyme deficiency that is autosomal dominant; most enzyme deficiencies are recessive
 - AIP is dominant because of high cellular demand for heme

As we progress through BPH1 and 2, and then into year 2 we will return to many of these disorders as we illustrate other genetic principles

A 5-year-old girl is diagnosed with neurofibromatosis and she and her family are treated accordingly. Which of the following is this child most likely at increased risk for as she ages?

- | | |
|--|-----|
| A. Pancreatitis | 20% |
| B. Chronic obstructive pulmonary disease | 20% |
| C. Broken bones with minimal trauma | 20% |
| D. Muscle wasting | 20% |
| E. Cancer | 20% |

Reading - In chapter 10 in the Korf textbook

A 5-year-old girl is brought to the physician because of progressive formation of neurofibromatosis. Genetic testing shows that she has a pathogenic variant in the *NF1* gene which is not found in either parent. Which of the following is this child most likely at increased risk for as she ages?

- | | |
|---|----|
| A. Pancreatitis | 0% |
| B. Chronic obstructive pulmonary disease | 0% |
| C. Broken bones with minimal trauma | 0% |
| D. Muscle wasting | 0% |
|  E. Cancer | 0% |

A 2-year-old girl is brought to the physician for follow-up examination because of global developmental delay. Genetic testing shows that a pathogenic variant in the *RNU4-2* gene which encodes snRNA which is a component of the spliceosome. There is no family history of the disorder. Which of the following best explains the inheritance of this disorder?

- | | |
|------------------------|----|
| A. Dominant negative | 0% |
| B. New mutation | 0% |
| C. X-linkage | 0% |
| D. Autosomal recessive | 0% |
| E. Gain of function | 0% |

A 2-year-old girl is brought to the physician for follow-up examination because of global developmental delay. Genetic testing shows that a pathogenic variant in the *RNU4-2* gene which encodes snRNA which is a component of the spliceosome. There is no family history of the disorder. Which of the following best explains the inheritance of this disorder?

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| C. X-linkage | 0% |
| D. Autosomal recessive | 0% |
| E. Gain of function | 0% |